LAMMPS official dem01-legacy-hooke-twosphere

Normal collision: restitution and momentum | 29 October 2026

PASS	Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted.

Official quantities, errors and acceptance limits

Quantity	Reference / limit	CPU dt	GPU dt	CPU dt/2	GPU dt/2
Total x momentum	0 absolute limit 3.534e-06	0 err 0 PASS	0 err 0 PASS	0 err 0 PASS	0 err 0 PASS
Restitution	0.49877 relative limit 0.005	0.4985282 err 0.0004848 PASS	0.4985282 err 0.0004848 PASS	0.4987053 err 0.0001298 PASS	0.4987053 err 0.0001298 PASS

Check time=8.385e-05 s; original step 2795. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

Original bodies and source-to-native mapping

Body	Radius / density	Initial xyz (m)	Initial v (m/s)	Initial spin (rad/s)
left	R=0.0025 rho=2700	(-0.00255, 0, 0)	(2, 0, 0)	(0, 0, 0)
right	R=0.0025 rho=2700	(0.00255, 0, 0)	(-2, 0, 0)	(0, 0, 0)

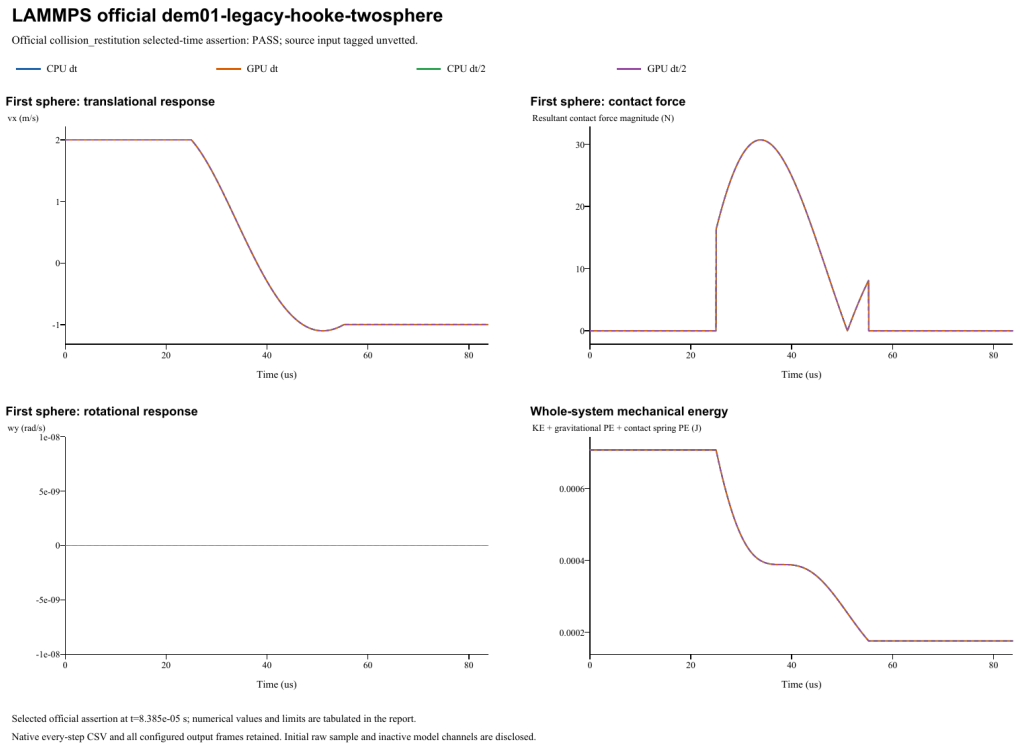
Linear Hooke Fn_elastic=kn*delta, kn=1000000 N/m. Manual c_n=4.064435496 N s/m, c_t=0 N s/m; automatic restitution damping is inactive. Tangential inputs: kt=285714.3 N/m (explicit in linear mode), mu=0. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0.

Gravity=(0, 0, 0) m/s^2. Base/half dt=3e-08/1.5e-08 s; 2795/5590 steps. Legacy normal cn=meff*gnorm exactly. NULL kt=2kn/7; NULL gt=gnorm/2 when tangential dampflag1, else0. Explicit native constants preserve these coefficients; tangential relative velocity is zero.

Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob fcb49be715ccc2b5efd6ed6c005c33d47256842c. Official tag: unvetted.
LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

LAMMPS native histories and refinement



CPU/GPU and half-step comparison

Quantity	CPU dt error	CPU dt/2 error	Error type / limit
Total x momentum	0	0	absolute: 3.534e-06
Restitution	0.00048478	0.00012981	relative: 0.005

CPU/GPU whole-history maximum position / velocity / spin / force differences: 0 m / 2.82e-17 m/s / 0 rad/s / 3.55e-15 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 2.6e-18 kg m/s / 2.17e-19 m / 2.18e-16 m/s. Gravity is added separately to saved native contact forces.

Complete results and verified scope

All 16,774 native frames are retained; main frame counts=[2796, 2796, 5591, 5591]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.